SQL Practical Assignment: AutoCare Workshop & Fleet Service Analytics

Topic: Table Constraints, DDL/DML & Data Analysis | Database: MySQL 8.0+ | Total Marks: 50

Scenario: You are appointed as a Database Analyst for **Apex AutoCare Hub**, a multi-city automotive service network across Maharashtra. The executive team wants a robust, normalized database system to monitor repair jobs, spare parts billing, mechanic assignments, and branch metrics while enforcing strict database integrity constraints.

TASK 1: DATABASE SETUP (4 MARKS)

1. Create database named `Apex_AutoCare_DB`.
2. Switch to `Apex_AutoCare_DB` for active operations.
3. Show all databases available on the MySQL server instance.
4. Verify active selection using `SELECT DATABASE();`.

ENFORCED INTEGRITY CONSTRAINTS

1. `PRIMARY KEY` on `Service_ID` (Unique & Non-null).
2. `NOT NULL` on Customer, Vehicle, & Branch.
3. `CHECK` `Quantity > 0` and `Labor_Cost >= 0`.
4. `DEFAULT` `Status = 'Completed'` & `Payment_Mode = 'UPI'`.

TASK 2: CREATE TABLE WITH CONSTRAINTS (6 MARKS)

Column Name	Data Type	Applied Constraints	Description
`Service_ID`	`INT`	`PRIMARY KEY`	Unique job card/invoice identifier
`Customer_Name`	`VARCHAR(50)`	`NOT NULL`	Full name of the vehicle owner
`Vehicle_Model`	`VARCHAR(40)`	`NOT NULL`	Car/SUV model serviced (e.g., Thar, Creta, Swift)
`Service_Type`	`VARCHAR(30)`	`NOT NULL`	Maintenance, Engine Overhaul, Brake Service, Electrical
`Parts_Replaced`	`INT`	`CHECK (Parts_Replaced >= 0)`	Total count of spare parts installed
`Total_Bill`	`DECIMAL(10,2)`	`CHECK (Total_Bill > 0)`	Final invoice amount in INR (₹)
`Branch_City`	`VARCHAR(35)`	`NOT NULL`	Workshop branch location
`Payment_Mode`	`VARCHAR(20)`	`DEFAULT 'UPI'`	UPI, Credit Card, Cash, Net Banking
`Lead_Mechanic`	`VARCHAR(40)`	`NOT NULL`	Assigned certified technician

TASK 3: INSERT ALL 20 RECORDS (DATASET INGESTION)

```
INSERT INTO Services (Service_ID, Customer_Name, Vehicle_Model, Service_Type, Parts_Replaced, Total_Bill, Branch_City,
Payment_Mode, Lead_Mechanic) VALUES
(201, 'Nilesh Deshmukh', 'Mahindra Thar', 'Engine Overhaul', 5, 28500.00, 'Ahilyanagar', 'Credit Card', 'Tanmay'),
(202, 'Priyanka Shinde', 'Hyundai Creta', 'Periodic Maintenance', 3, 6200.00, 'Pune', 'UPI', 'Santosh'),
(203, 'Sanket Patil', 'Tata Nexon EV', 'Electrical Diagnostics', 1, 4800.00, 'Mumbai', 'UPI', 'Rajesh'),
(204, 'Aniket Jadhav', 'Maruti Swift', 'Brake Pad Replacement', 2, 3400.00, 'Nashik', 'Cash', 'Ganesh'),
(205, 'Kavita Kadam', 'Toyota Fortuner', 'Suspension Overhaul', 6, 42000.00, 'Ahilyanagar', 'Net Banking', 'Tanmay'),
(206, 'Mahesh Shirole', 'Kia Seltos', 'Periodic Maintenance', 2, 5800.00, 'Pune', 'UPI', 'Santosh'),
(207, 'Shubham More', 'Mahindra Scorpio-N', 'Engine Overhaul', 4, 24500.00, 'Chhatrapati Sambhajinagar', 'Credit Card',
'Rajesh'),
(208, 'Sneha Bhalerao', 'Honda City', 'Air Conditioning Service', 2, 7500.00, 'Mumbai', 'UPI', 'Ganesh'),
(209, 'Amol Gaikwad', 'Maruti Brezza', 'Periodic Maintenance', 3, 5200.00, 'Nashik', 'UPI', 'Santosh'),
(210, 'Vaishali Pawar', 'Toyota Innova Crysta', 'Transmission Repair', 7, 36000.00, 'Pune', 'Credit Card', 'Tanmay'),
(211, 'Rohan Jagtap', 'Tata Punch', 'Electrical Diagnostics', 1, 2900.00, 'Ahilyanagar', 'Cash', 'Rajesh'),
(212, 'Pooja Kulkarni', 'Hyundai Venue', 'Brake Pad Replacement', 2, 3800.00, 'Pune', 'UPI', 'Ganesh'),
(213, 'Deepak Salunke', 'Mahindra Thar', 'Periodic Maintenance', 4, 8500.00, 'Chhatrapati Sambhajinagar', 'UPI', 'Santosh'),
(214, 'Pallavi Joshi', 'Maruti Swift', 'Clutch Replacement', 3, 9200.00, 'Mumbai', 'Credit Card', 'Tanmay'),
(215, 'Suresh Sonawane', 'Toyota Fortuner', 'Engine Overhaul', 5, 31000.00, 'Nashik', 'Net Banking', 'Rajesh'),
(216, 'Archana Thorat', 'Tata Nexon', 'Periodic Maintenance', 2, 4900.00, 'Ahilyanagar', 'UPI', 'Santosh'),
(217, 'Gaurav Kale', 'Kia Seltos', 'Air Conditioning Service', 2, 7100.00, 'Pune', 'UPI', 'Ganesh'),
(218, 'Meera Wagh', 'Hyundai Creta', 'Suspension Overhaul', 4, 18500.00, 'Chhatrapati Sambhajinagar', 'Credit Card', 'Tanmay'),
(219, 'Prashant Bhise', 'Honda Elevate', 'Electrical Diagnostics', 1, 3100.00, 'Mumbai', 'UPI', 'Rajesh'),
(220, 'Tejas Nalawade', 'Mahindra Scorpio-N', 'Periodic Maintenance', 3, 7600.00, 'Ahilyanagar', 'Cash', 'Ganesh');
```

TASK 4: FILTERING & SELECTION (6 MARKS)

1. Display all service records from the `Services` table.
2. Display only `Customer_Name`, `Vehicle_Model`, and `Total_Bill`.
3. Find all repair jobs handled at the `Ahilyanagar` workshop.
4. Display all jobs categorized under `Engine Overhaul`.
5. List all invoices where `Total_Bill` exceeds ₹15,000.
6. List records where `Parts_Replaced` is 4 or more.
7. Find all repairs supervised by mechanic `Tanmay`.
8. Display jobs where `Total_Bill` is between ₹5,000 and ₹20,000.
9. Display jobs located in either `Pune` or `Mumbai`.
10. Find vehicle models containing the keyword `'Thar'` or `'Scorpio'`.
11. List all service jobs sorted by `Total_Bill` in descending order.
12. Display the top 5 highest-grossing repair invoices.

TASK 5 & 6: DDL, UPDATE & DELETE (6 MARKS)

1. **UPDATE:** Change payment mode of `Service_ID` 204 to `'UPI'`.
2. **UPDATE:** Increase `Total_Bill` by 8% for all `Suspension Overhaul` jobs.
3. **UPDATE:** Reassign all jobs under mechanic `'Ganesh'` in Ahilyanagar to `'Tanmay'`.
4. **ALTER TABLE:** Add a constraint checking that `Parts_Replaced <= 25`.
5. **DELETE:** Delete the repair record with `Service_ID = 211`.
6. **DELETE:** Remove all service records where payment was made in `'Cash'`.
7. **DELETE:** Delete all records where `Total_Bill < 3000.00`.

Apex AutoCare Hub: Analytical Aggregations & Decision Support Queries

Section 2: Multi-Dimensional Aggregation, Filtering Groups, and Executive Metrics

TASK 7: AGGREGATE FUNCTIONS (6 MARKS)

1. `COUNT()` – Find the total number of service jobs logged.
2. `SUM()` – Calculate the grand total revenue generated across all jobs.
3. `AVG()` – Compute the overall average invoice bill.
4. `MAX()` – Find the most expensive repair bill in the database.
5. `MIN()` – Find the lowest repair bill logged.
6. `COUNT()` – Total number of services paid via `UPI`.
7. `SUM()` – Total spare parts replaced in the Pune branch.
8. `AVG()` – Average billing amount for `Periodic Maintenance`.
9. `MAX()` – Highest number of spare parts replaced in a single job.
10. `MIN()` – Minimum bill for `Engine Overhaul` services.

TASK 8: GROUP BY OPERATIONS (8 MARKS)

1. Find the total revenue earned by each `Branch_City`.
2. Count the number of service jobs handled by each `Lead_Mechanic`.
3. Calculate the average bill for each `Service_Type`.
4. Find the total parts replaced grouped by `Vehicle_Model`.
5. Find the maximum repair bill logged for each `Service_Type`.
6. Find the minimum repair bill recorded in each `Branch_City`.
7. Count the total jobs completed under each `Payment_Mode`.
8. Calculate average parts replaced per `Service_Type`.
9. Find total revenue grouped by both `Branch_City` and `Service_Type`.
10. Count service records grouped by `Lead_Mechanic` and `Branch_City`.

TASK 9: GROUP BY WITH HAVING (8 MARKS)

1. Display cities where total revenue generated exceeds ₹40,000.
2. Display service types that have been performed more than 3 times.
3. Display mechanics who handled more than 4 service jobs.
4. Display service types having an average bill greater than ₹10,000.
5. Display vehicle models with total repair bills exceeding ₹30,000.
6. Display branches where total parts replaced exceed 8 units.
7. Display payment modes accounting for more than 5 transactions.
8. Display mechanics whose total supervised billing exceeds ₹50,000.
9. Display cities with an average repair bill greater than ₹12,000.
10. Display service types where the maximum single bill exceeds ₹25,000.

TASK 10: BUSINESS INTELLIGENCE & INSIGHTS (6 MARKS)

1. **Highest Revenue Vehicle:** Which vehicle model generated the highest cumulative billing?
2. **Top Producing Mechanic:** Which mechanic brought in the maximum total revenue for the company?
3. **Branch Performance:** Which city branch completed the most service jobs?
4. **Preferred Transaction Gateway:** Which payment method generated the highest total cashflow?
5. **Complex Jobs:** Which service category requires the highest average number of parts replaced?
6. **High-Value Customers:** Retrieve all jobs where `Total_Bill > 25000` along with customer name and vehicle.
7. **Operational Efficiency:** Calculate average billing per part replaced across each service type.
8. **Volume Analysis:** List all branches where total jobs logged are at least 4.
9. **Spare Parts Consumption:** Find the overall sum of parts consumed across the entire network.
10. **Profit Optimization:** Rank the top 3 most profitable services based on average billing value.

Exam Submission Checklist: Ensure all DDL statements include `PRIMARY KEY`, `NOT NULL`, `CHECK`, and `DEFAULT` clauses. Test SQL script execution sequentially in MySQL Workbench or Command Line Client.